1. Fraction of Violations in terms of flipping rates

Let $p = P(x_i = 1)$. As a first approximation,

$$P(00) = (1-p)^2, \qquad P(10) = 2p(1-p), \qquad P(11) = p^2,$$

where $P(ij)$ is the probability that a new clause attaches to variables that are in the i and j state. Now let F_{ij} be the probability that a proposed flip to a (i,j) pair is accepted. Note that if a pair $(1,1)$ is flipped, the number of ones decreases by two. If a pair $(0,0)$ is flipped, the number of ones increases by two. If a pair $(1,0)$ is flipped, the number of ones doesn't change. So the rate equation for the number of ones is

$$\frac{dp}{dt} = 2(1-p)^2 F_{00} - 2p^2 F_{11}. \tag{1}$$

Before finding F_{00} and F_{11}, let's see what we can do with this. At steady state,

$$0 = 2(1-p)^2 F_{00} - 2p^2 F_{11}.$$

Solving for the equilibrium value $\bar{p}$, we get

$$\bar{p} = \frac{\sqrt{F_{00}}}{\sqrt{F_{00}} + \sqrt{F_{11}}}.$$

Now we estimate the number of violations. For violations of AND clauses, the two states must not be ones, so the probability of an AND clause being violated is

$$V_{\text{AND}} = 1 - p^2.$$

Similarly, the probability of an XOR clause being violated is

$$V_{\text{XOR}} = p^2 + (1-p)^2.$$

Therefore, in the case where we have equal AND and XOR clauses, the fraction of violations is

$$V = \frac{1}{2}\left[1 - \bar{p}^2\right] + \frac{1}{2}\left[\bar{p}^2 + (1-\bar{p})^2\right]. \tag{2}$$

So, if we can estimate F_{00} and F_{11}, we can estimate the number of violations in steady state (and possibly estimate its time course). So let's focus on estimating these rates.

2. Estimation of flipping rates

Consider that a clause attaches to a pair $(0,0)$, and consider what happens if the pair is flipped to $(1,1)$. Let N^+ be the number of newly created violations and N^- the number of newly removed violations. Since $M = \alpha K$ and each clause has two variables, this means that each variable appears, on average, in 2α clauses. For equally distributed AND and XOR clauses, then, each variable appears in α XOR clauses and α AND clauses. Look at one of the variables that is flipped from 0 to 1. It has α AND clauses attached to it. Of those, αp are ones, so these pairs flip as $(0,1) \to (1,1)$, and all these are new removed violations. Similarly, there are $\alpha(1-p)$ zeros, and these pairs flip as $(0,0) \to (1,0)$, which doesn't change the violations. In summary, violations from neighboring AND clauses have mean $N^- = \alpha p$. Now consider the α XOR clauses. Of those, αp flip from $(0,1)$ to $(1,1)$ creating violations. The others are $\alpha(1-p)$ that flip from $(0,0)$ to $(1,0)$, removing violations. In summary, neighboring XOR clauses produce $N^+ = \alpha p$ new violations and remove $N^- = \alpha(1-p)$ violations.

So, for one of the flipped zeros we have

$$E[N^+] = \alpha p,$$
$$E[N^-] = \alpha p + \alpha(1-p) = \alpha.$$

Taking into account the two zeros that are flipped,

$$E[N^+] = 2\alpha p,$$
$$E[N^-] = 2\alpha.$$

Now, these are the averages, but the actual number of violations is random. We can assume that the distribution is Poisson with the averages above,

$$N^+ = \text{Poisson}(2\alpha p),$$
$$N^- = \text{Poisson}(2\alpha),$$

With these, we can get the distribution of the *net* change in the number of violations from a $(0,0)$ flip:

$$Z_{00} = \text{Poisson}(2\alpha p) - \text{Poisson}(2\alpha)$$

Doing a similar argument for flipping a $(1,1)$ pair to $(0,0)$ we get

$$Z_{11} = \text{Poisson}(2\alpha) - \text{Poisson}(2\alpha p)$$

Now consider the case where the incoming clause is AND. Define $H_{00}(k) = P(Z_{00} \leq k)$. The new AND clause on the flipped $(1,1)$ pair is satisfied, but we need to determine if the removed clause was satisfied or not. If it was satisfied as well, there is no net difference from the new clause and the clause will be accepted if the neighbor clauses decrease the violations, that is with probability $H_{00}(0)$. So

$$\begin{aligned}
&P(\text{AND clause attached to 00 accepted}) = \\
&P(\text{removed clause was violated})H_{00}(1) + P(\text{removed clause was satisfied})H_{00}(0) \\
&= (1-\bar{p}^2)H_{00}(1) + \bar{p}^2 H_{00}(0).
\end{aligned}$$

Similarly, for an XOR clause, it automatically comes with a violation since it is attaching to a $(0,0)$ pair, so one violation more needs to be removed to accept the pair

$$\begin{aligned}
&P(\text{XOR clause attached to 00 accepted}) = \\
&P(\text{removed clause was violated})H_{00}(0) + P(\text{removed clause was satisfied})H_{00}(-1) \\
&= [\bar{p}^2 + (1-\bar{p})^2]H_{00}(0) + [1 - \bar{p}^2 - (1-\bar{p})^2]H_{00}(-1).
\end{aligned}$$

Since half the incoming clauses are AND and half are XOR, then

$$\begin{aligned}
F_{00} = &= \frac{1}{2}\left([\bar{p}^2 + (1-\bar{p})^2]H_{00}(0) + [1 - \bar{p}^2 - (1-\bar{p})^2]H_{00}(-1)\right) \\
&+ \frac{1}{2}\left((1-\bar{p}^2)H_{00}(1) + \bar{p}^2 H_{00}(0)\right)
\end{aligned}$$

Doing the same argument for a new clause attaching to a $(1,1)$ pair, we get

$$F_{11} = \frac{1}{2}\left[(1-\bar{p}^2)H_{11}(0) + \bar{p}^2 H_{11}(-1)\right] + \frac{1}{2}\left[\left(\bar{p}^2 + (1-\bar{p})^2\right)H_{11}(0) + 2\bar{p}(1-\bar{p})H_{11}(-1)\right].$$

3. Numerical Test

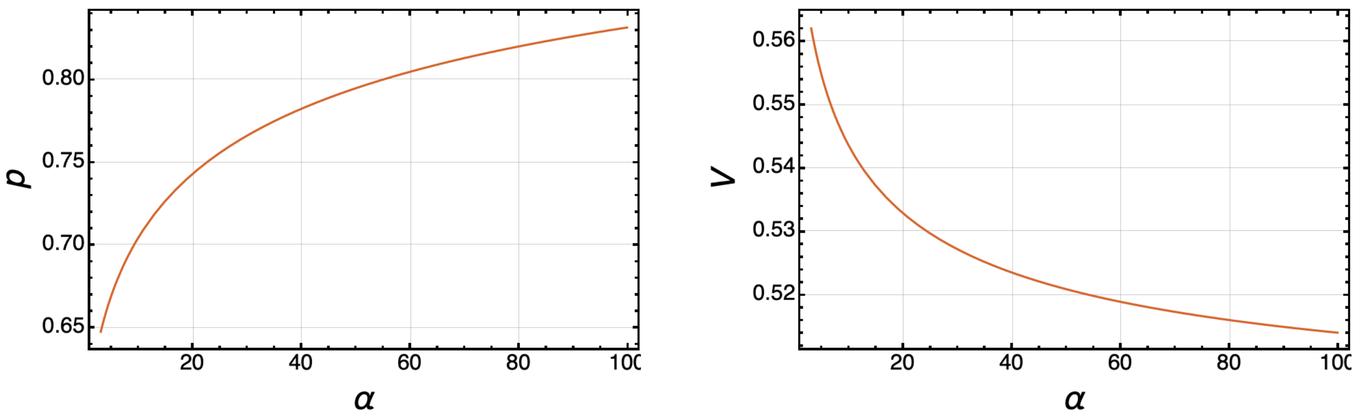
To find the fraction of ones p^*, we need to solve the following implicit equaiton for p^*

$$\frac{\bar{p}^2}{(1-\bar{p})^2} = \frac{F_{00}}{F_{11}} = \frac{[\bar{p}^2 + (1-\bar{p})^2]H_{00}(0) + [1-\bar{p}^2-(1-\bar{p})^2]H_{00}(-1) + (1-\bar{p}^2)H_{00}(1) + \bar{p}^2 H_{00}(0)}{(1-\bar{p}^2)H_{11}(0) + \bar{p}^2 H_{11}(-1) + \left(\bar{p}^2 + (1-\bar{p})^2\right)H_{11}(0) + 2\bar{p}(1-\bar{p})H_{11}(-1)}$$

Simplifying,

$$\frac{\bar{p}^2}{(1-\bar{p})^2} = \frac{F_{00}}{F_{11}} = \frac{[1 - 2\bar{p} + 3\bar{p}^2]H_{00}(0) + 2\bar{p}(1-\bar{p})H_{00}(-1) + (1-\bar{p}^2)H_{00}(1)}{(2 - 2\bar{p} + \bar{p}^2)H_{11}(0) + \bar{p}(2-\bar{p})H_{11}(-1)}$$

Solving this equation numerically we can find $\bar{p}$, and then we can plug it into Eq. (1) to find the fraction of violations. Doing this we can plot p^* and V as a function of α, as shown in the figures below. The fraction of ones $\bar{p}$ converges to 1, the mean-field value, when $\alpha \to \infty$ (not shown).



p
0.80
0.75
0.70
0.65
20
40
60
80
100
α
V
0.56
0.55
0.54
0.53
0.52
20
40
60
80
100
α